



Security of Mobile Networks

PROF. TEWFIQ EL MALIKI

2025

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Cellular Networks Introduction

SECURITY

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Evolution of Cellular Networks

Mobile 1G

AMPS, NMT, TACS



Mobile 2G

D-AMPS, GSM/GPRS,
cdmaOne

Mobile 3G

CDMA2000/EV-DO,
WCDMA/HSPA+, TD-SCDMA

Mobile 4G LTE

LTE, LTE Advanced



N/A

Analog Voice

<0.5 Mbps¹

Digital Voice + Simple Data

63+ Mbps²

Mobile Broadband

300+ Mbps³

Faster and Better



4 | Security of 1G AMPS (Advanced Mobile Phone Service)

- **No security!**
- Used a combination of Electronic Serial Number and Cellular Telephone Number to identify itself on the network
- All communications (control and voice) were done using analog tones
- Eavesdropping
- Mobile Cloning



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GSM

- GSM (Global System for Mobile Communications) is still the most widely used cellular standard
 - > 600 million users, mostly in Europe and Asia; limited coverage and support in USA
 - Based on TDMA radio access and PCM trunking
 - Use SS7 signalling with mobile-specific extensions
 - Provides authentication and encryption capabilities
 - Third generation (3G) and future (4G)

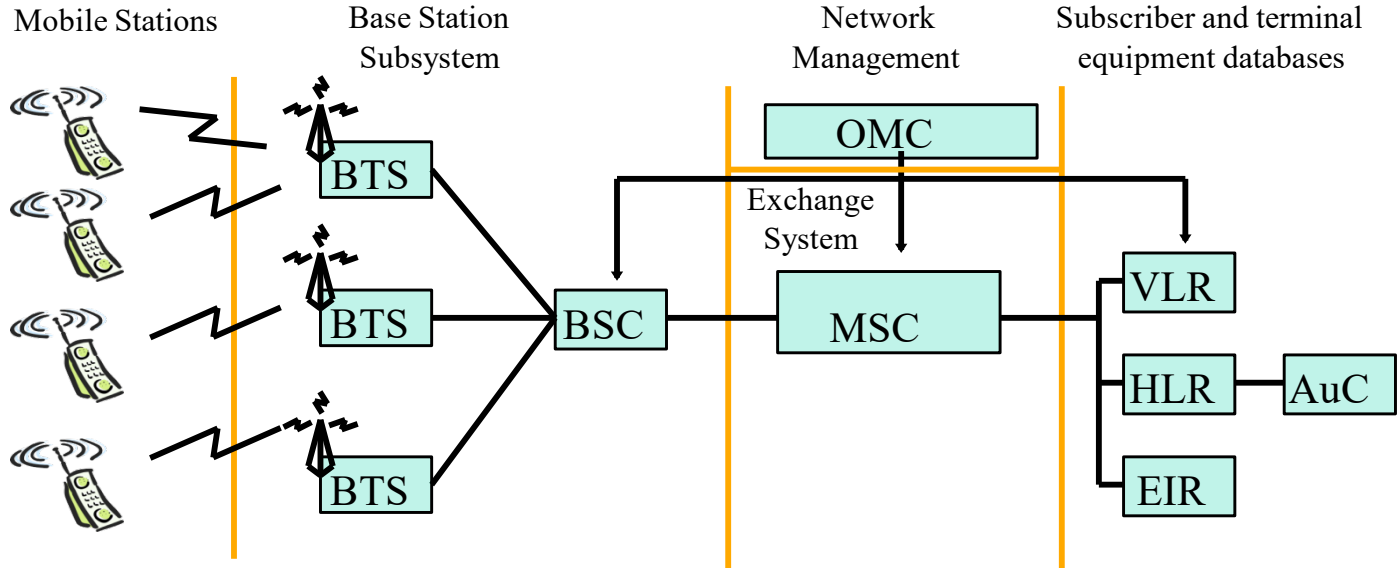
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GSM

- 900 MHz (or 1800 MHz) band
 - uplink frequency band 890-915 MHz
 - downlink frequency band is 935-960 MHz
 - 25 MHz subdivided into 124 carrier frequency channels, each 200 kHz apart
- Time division multiplexing (TDMA)
 - allows 8 speech channels per radio frequency channel
 - Channel data rate is 270.833 kbps
 - Voice transmitted at 13 kbps débit téléphone fixe $D=64 \text{ kbit/s} = 8000 * 8 \text{ bits}$
 - Handset power max. 2 watts in GSM850/900 and 1 watt in GSM1800/1900
 - Cell size up to 35 km
- $A(\text{LOS}) = 32 + 20 \text{ LOG}(\text{dkm}) + 20 \text{ LOG}(\text{fMHz})$ $k = 1,38 \text{ E}(-23) \text{ w}/(^{\circ}\text{K} \cdot \text{Hz})$
- $30 \text{ dBm} - (-100) = 32 + 20 \text{ LOG}(\text{dkm}) + 20 \text{ LOG}(1000) \rightarrow 130 - 32 - 20 * 3 = 20 \text{ log}(\text{dkm}) \rightarrow 2 = \text{log}(\text{dkm}) \rightarrow d = 100 \text{ km}$

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GSM Architecture



HLR = Home Location Register AC
= Authentication center

VLR = Visitor Location Register **EIR -**
Equipment Identity Register

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Goals

Confidentiality and Anonymity on the radio path

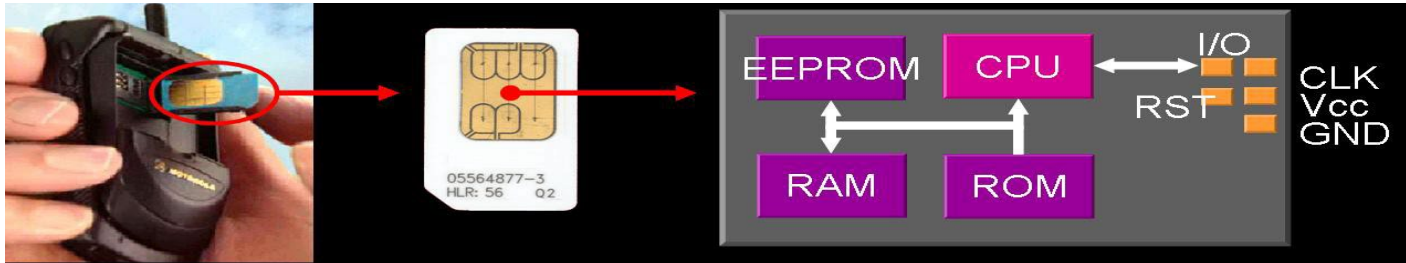
Strong client authentication to protect the operator against the billing fraud

- Prevention of operators from compromising of each others' security
 - Inadvertently
 - Competition pressure

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SIM (Subscriber Identification Module)

- Subscriber Identification Module (SIM)
 - Smart Card – a single chip computer containing OS, File System, Applications
 - Owned by operator (*i.e. trusted*)



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SIM Cards

Typical specification

- 8 bit CPU
- 16 K ROM
- 256 bytes RAM
- 4K EEPROM
- Cost: \$5

Smart Card Technology

- Based on ISO 7816 defining
- Card size, contact layout, electrical characteristics
- I/O Protocols: byte/block based
- File Structure

GSM Mobile : Mobile Equipment (ME)

- Physical mobile device Identifiers
- IMEI – International Mobile Equipment Identity Subscriber Identity Module (SIM)
- Smart Card containing keys, identifiers and algorithms Identifiers
 - **Ki** – Subscriber Authentication Key
 - **IMSI** – International Mobile Subscriber Identity
 - **TMSI** – Temporary Mobile Subscriber Identity
 - **MSISDN** – Mobile Station International Service Digital Network
 - **PIN** – Personal Identity Number protecting a SIM
 - **LAI** – location area identity

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Les identités

IMSI = MCC +MNC +NMSI

- MAX. 15 digits
- MCC (“Mobile Country Code”) : identifie le pays (i.e. PLMN)
- MNC (“Mobile Network Code”) :identifie l’opérateur
- NMSI (“Network Mobile Subscriber Identity”) : identifie l’abonné dans le réseau de l’opérateur.

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Les identités

MCC Pays MNC opérateurs Ecran

- 262 Allemagne 07 Viag Interkom D INTERKOM
- 228 Suisse 01 Swisscom CH NATEL D
- 228 Suisse 02 Sunrise CH SUNRISE
- 228 Suisse 03 Orange Comm. SA CH ORANGE

LAI (location area code)

- Ce code est diffusé régulièrement par la station de base sur le
BACH LAI = CC + MNC + LAC

Identifiants MSRN TMSI et LAI

Cell Identifier (CI)

- Il est en maximum de 2x8bits et identifie la cellule. le LAI+CI identifie d'une manière unique une cellule au niveau international

MSRN ("Mobile Station Roaming Number")

- Le numéro d'acheminement, conforme à E164 (mais non associé à un MS) qui permet aux commutateurs d'atteindre en cas d'appel vers un mobile qui se trouve dans un autre réseau (gestion d'itinérance), le MSC dans l'aire duquel se trouve le MS. Le MSRN est alloué par le VLR où se trouve le MS.

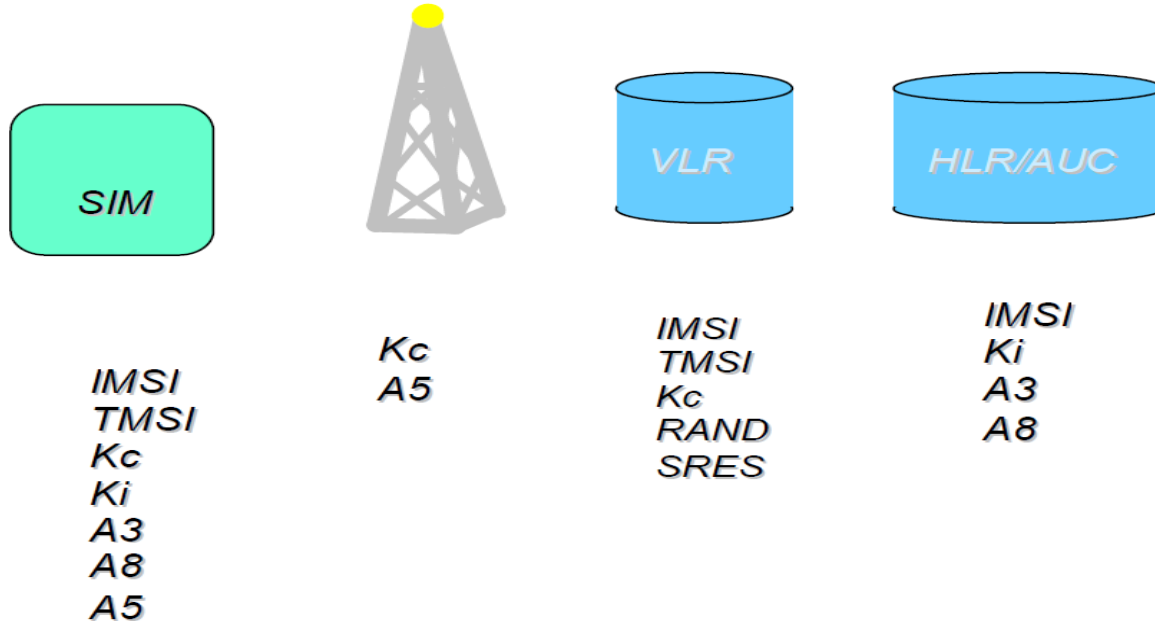
Identifiants MSRN TMSI et LAI

TMSI (“Temporary Mobile Subscriber Identity”)

- identité temporaire qui se substitue à l’IMSI et qui évite le transport de l’IMSI sur l’air – interface (confidentialité). Il est attribué par le VLR (4 bytes sauf FFFF)
- Le TMSI est associé à un IMSI et à une Location area où se trouve le MS Le couple (TMSI,LAI) remplace le IMSI et permet une identification unique.

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Information Repartition



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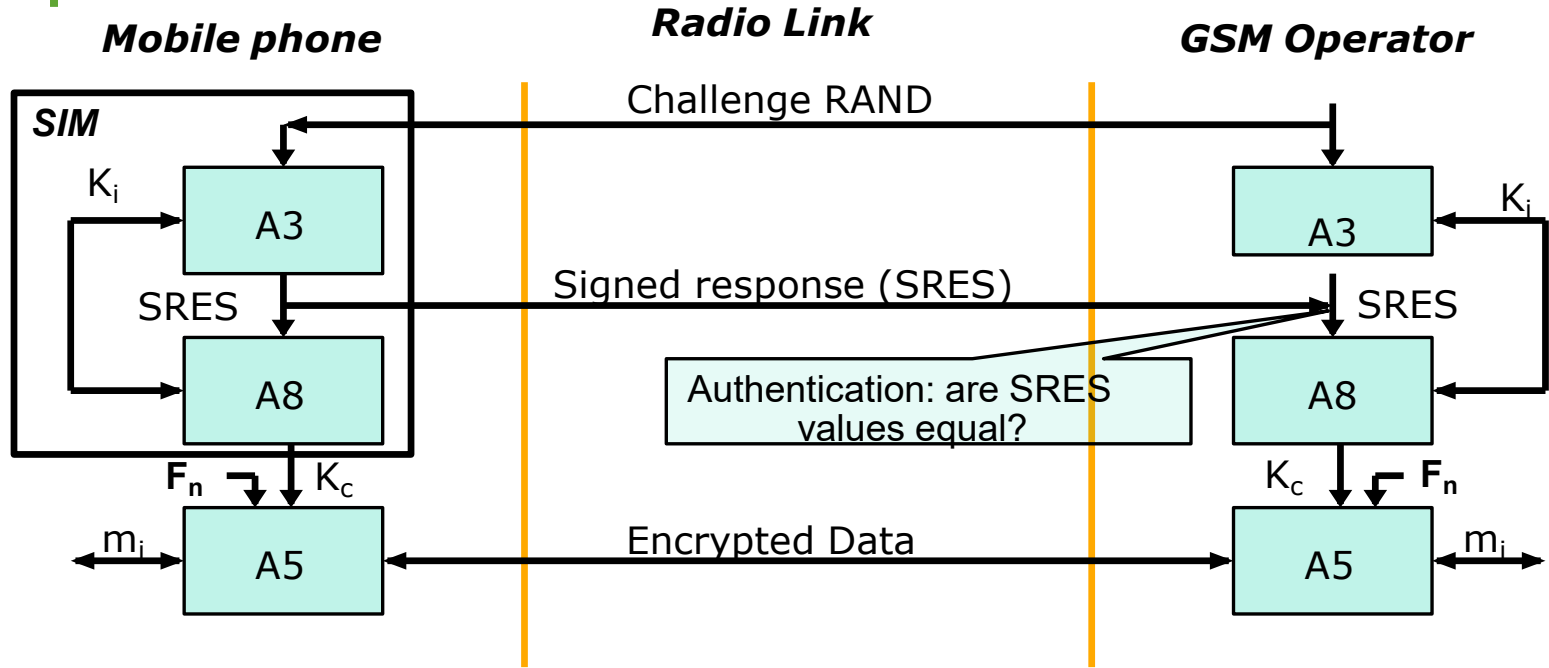
The Key is in the Card

- Ki – Subscriber Authentication Key
 - Shared 128 bit key used for authentication of subscriber by the operator
- Key Storage
 - Subscriber's SIM (owned by operator, i.e. trusted)
 - Operator's Home Locator Register (HLR) of the subscriber's home network



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Authentication



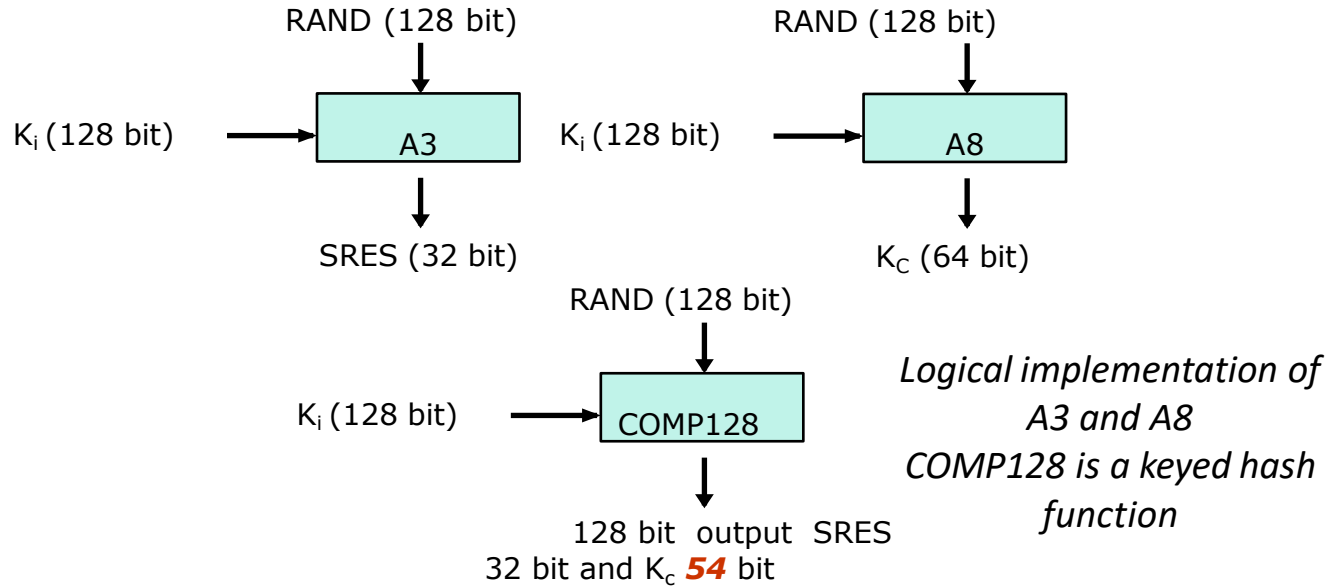
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A3 and A8 (Authentication and Session Key)

- Both A3 and A8 algorithms are implemented on the SIM
- Operator can decide, which algorithms to use.
- Algorithm implementation is independent of HW and operators.
- A8 was never made public

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A3 and A8 (Authentication and Session Key)



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A5 (Confidentiality)

A5 is a stream cipher

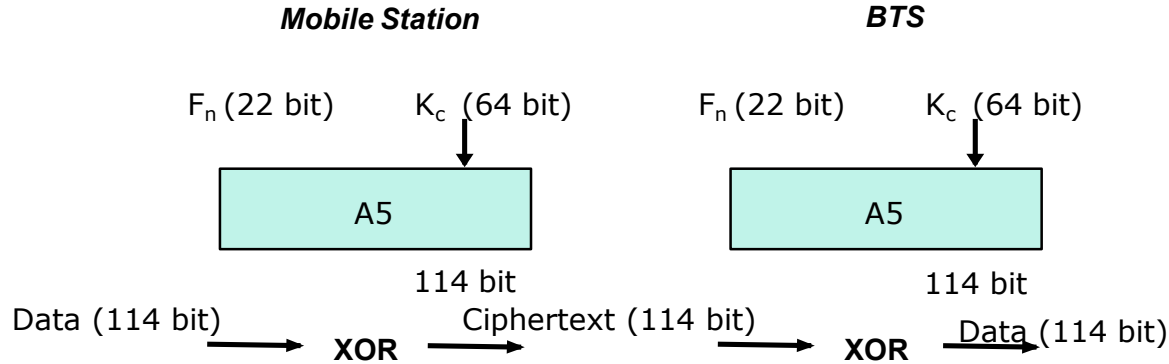
- Implemented very efficiently on hardware Design was never made public
- Leaked to Ross Anderson and Bruce Schneier

Variants: A5/1 – the strong version, A5/2 – the weak version, A5/3 GSM Association Security Group and 3GPP design

Based on Kasumi algorithm used in 3G mobile systems

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A5 (Confidentiality)



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Attack: Extracting the Key from the SIM card

Attack Goal

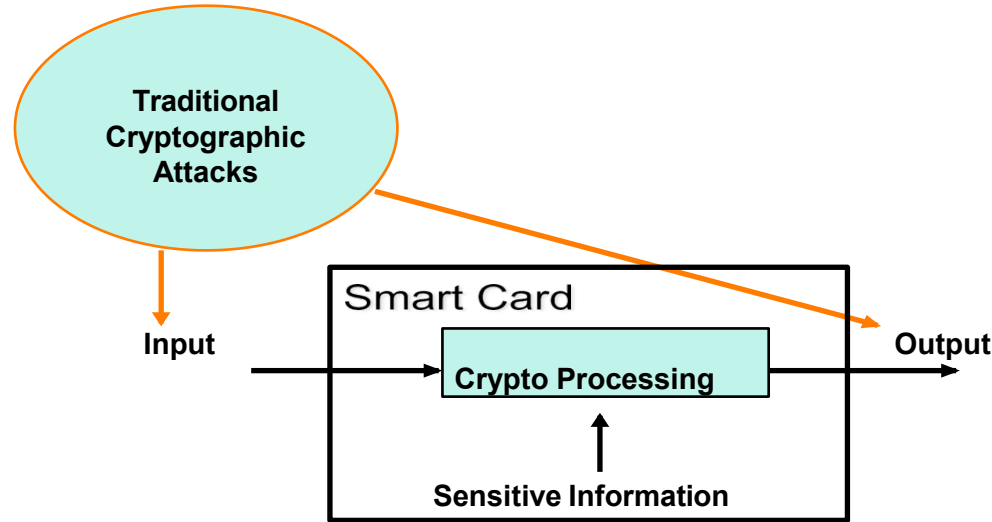
- Ki stored on SIM card
- Knowing Ki it's possible to clone SIM Cardinal
- Principle
 - Relevant bits of all intermediate cycles and their values should be statistically independent of the inputs, outputs, and sensitive information.

Attack Idea

- Find a violation of the Cardinal Principle, i.e. side channels with signals does depend on input, outputs and sensitive information
- Try to exploit the statistical dependency in signals to extract a sensitive information

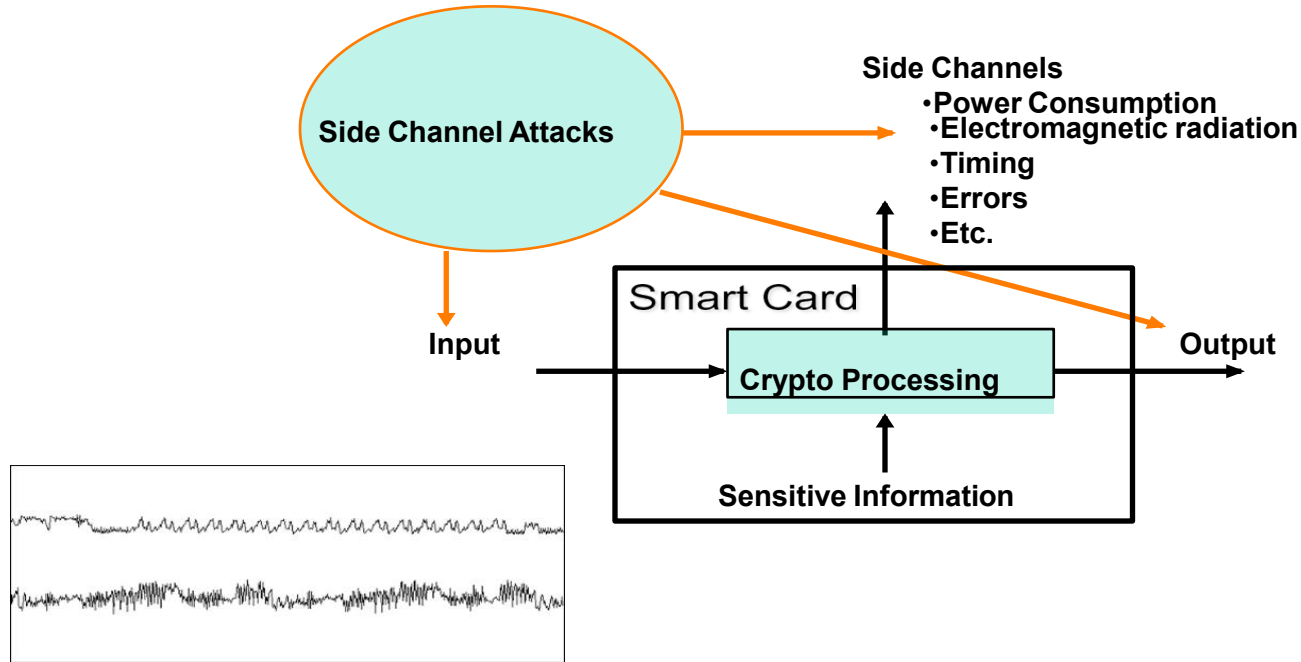
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Attack: Extracting the Key from the SIM card



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Attack: Extracting the Key from the SIM card



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Attack: Fake BS

- IMSI catcher by Law Enforcement
- Intercept mobile originated calls

tised to be ...

tiSRP, OpenBTS



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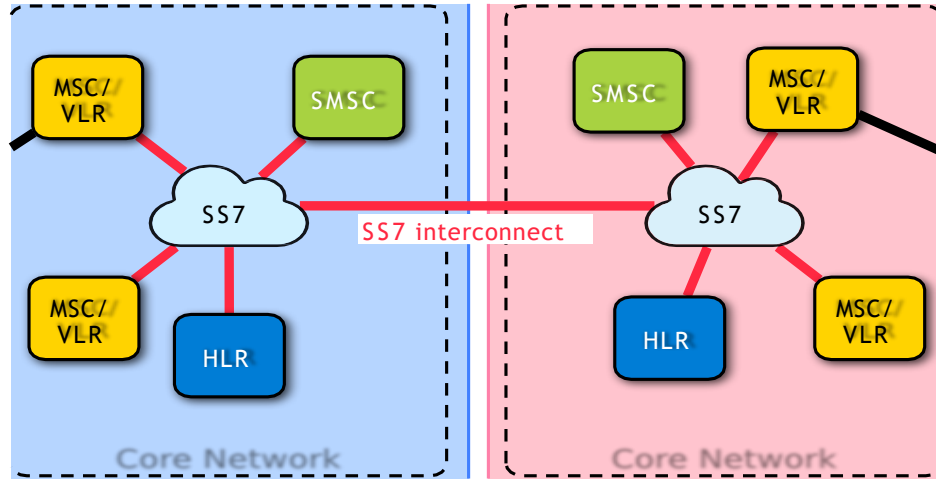
Signalling System #7

Attack: Location Tracking using SS7

- Signalling System #7 (SS7) is a protocol suite used by most telecommunication service providers to talk to each other
- Standardized in 1980's. **Trust model:** Service providers trust each other. **No authentication built in.**
- SS7 access can be bought from telecom providers for a few hundred dollars a month. Also, many unsecured SS7 hubs present on the web.

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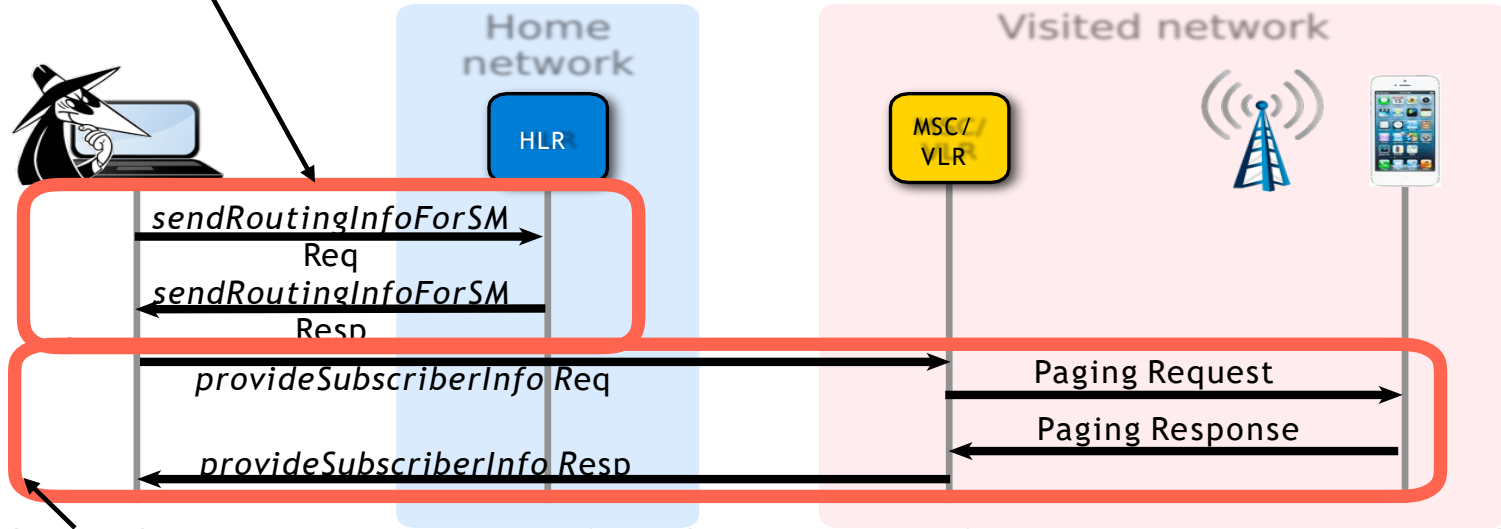
Attack: Location Tracking using SS7



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Attack: Location Tracking using SS7

Step 1: Get IMSI and address of current MSC



Step 2: Request the cell id of the subscriber to the current MSC

Several online services allow locating the subscriber using the paging response.

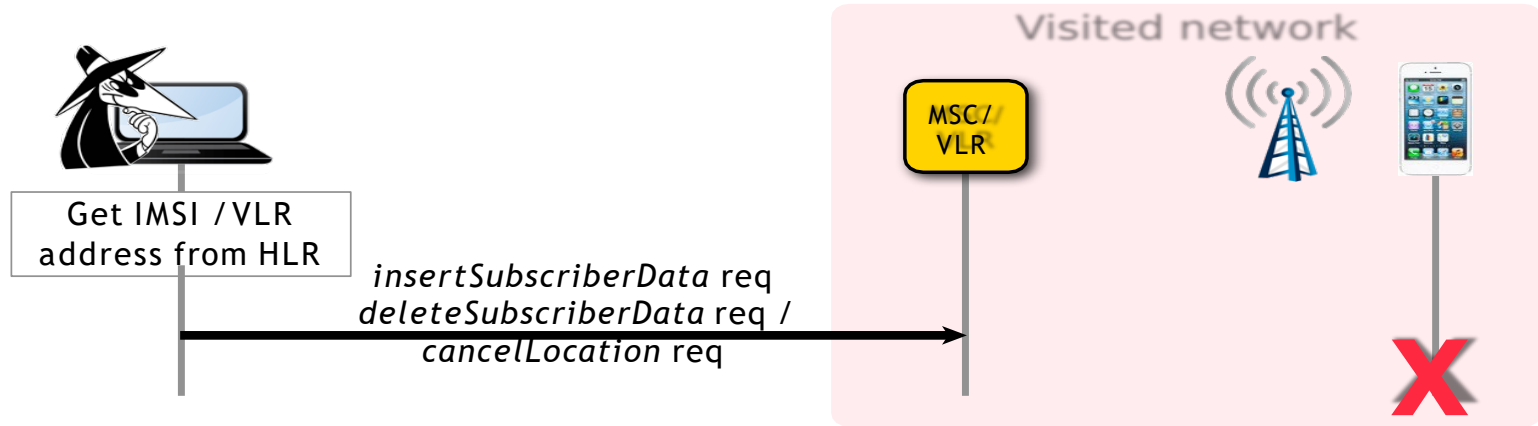
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SS7: Denial of Service

- Attacker can modify subscriber data as well.
- No checks implemented by most telecom providers.
- Once IMSI and VLR addresses are available to the attacker, he can control all kinds of service availability to the subscriber e.g., disabling outgoing calls etc.

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SS7: Denial of Service



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SS7 attacks

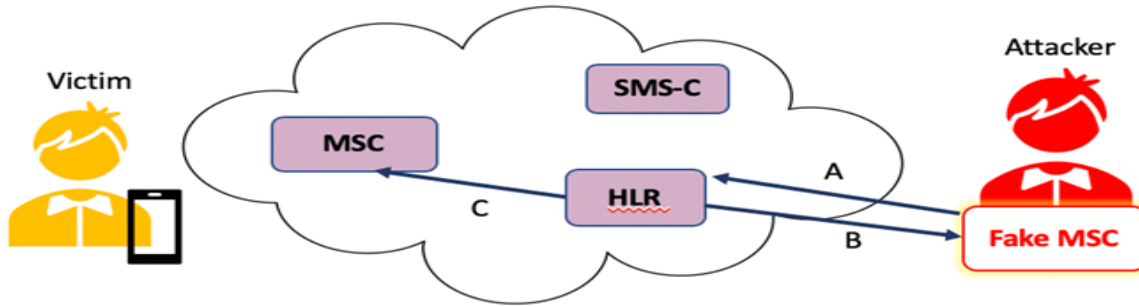
- Attacks that exploit security vulnerabilities in the SS7 protocol to compromise and intercept voice and SMS communications on a cellular network.
- It is similar to a Man In the Middle attack .

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SS7 attacks

- A. The attacker registers the Victim's MSISDN on the fake MSC
- B. The real HLR sets a new location for the Victim's MSISDN
- C. The real HLR requests the real MSC to release a memory

SMS Interception. Step 1: Set up

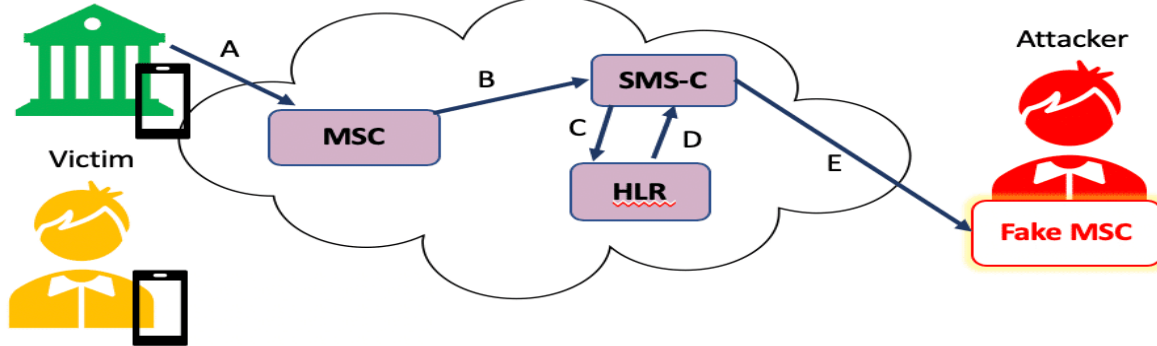


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SMS Attacks

SMS Interception. Step 2: Hijacking

Victim's Bank



- A. The Victim's bank send the Victim and SMS (e.g. a two-factor authentication code)
- B. The real MSC translates the SMS to the SMS-C
- C. The real SMS-C asks the real HLR for the Victim's location
- D. The real HLR replies with the MSC address, which is fake and controlled by the Attacker
- E. The real SMS-C translates the SMS to the fake MSC (the Attacker)

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Universal Mobile Telecommunications System (UMTS)

3G

UMTS Security

- Reuse of 2nd generation security principles (GSM):
 - Removable hardware security module
 - In GSM: SIM card
 - In 3GPP: USIM (User Services Identity Module)
 - Radio interface encryption
 - Limited trust in the Visited Network
 - Protection of the identity of the end user
 - Correction of the following weaknesses of the previous generation:
 - ATacks from a faked base station
 - Cipher keys and authentication data transmiTed in clear between and within networks
 - Encryption not used in some networks
 - Data integrity not provided

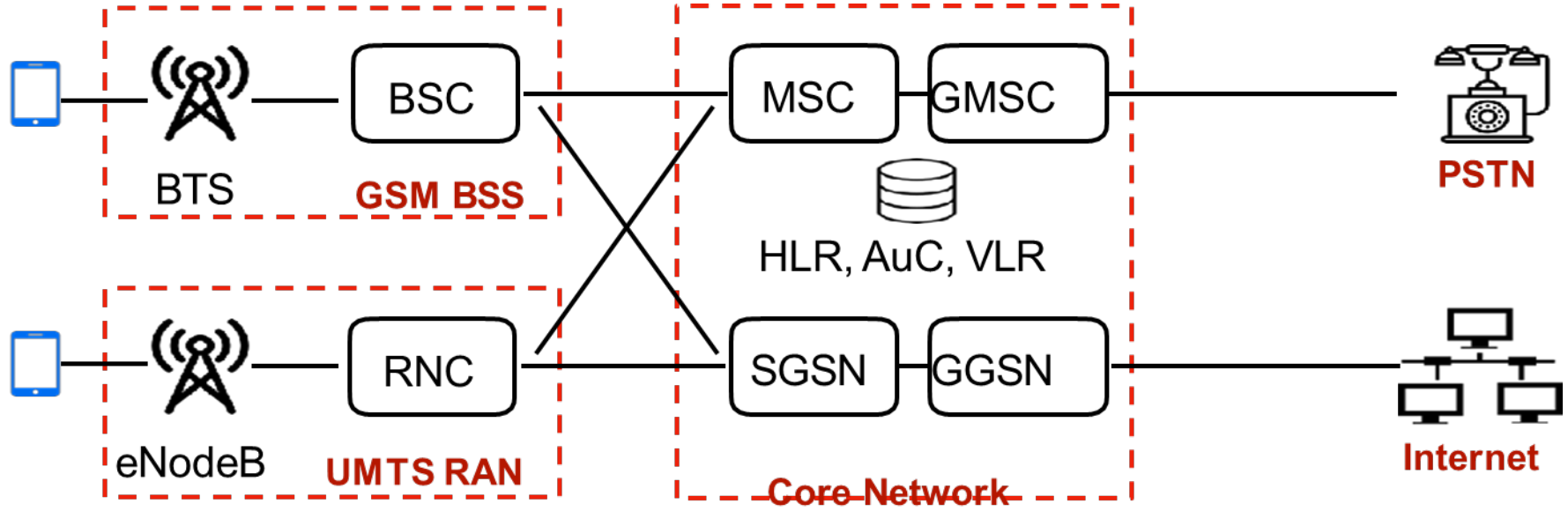
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GSM —> UMTS

- SIM —> USIM
- BTS —> eNodeB
- BSC —> RNC (Radio Network Controller)
- Backend (MSC, GMSC, SGSN, GGSN all remain the same)

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GSM → UMTS



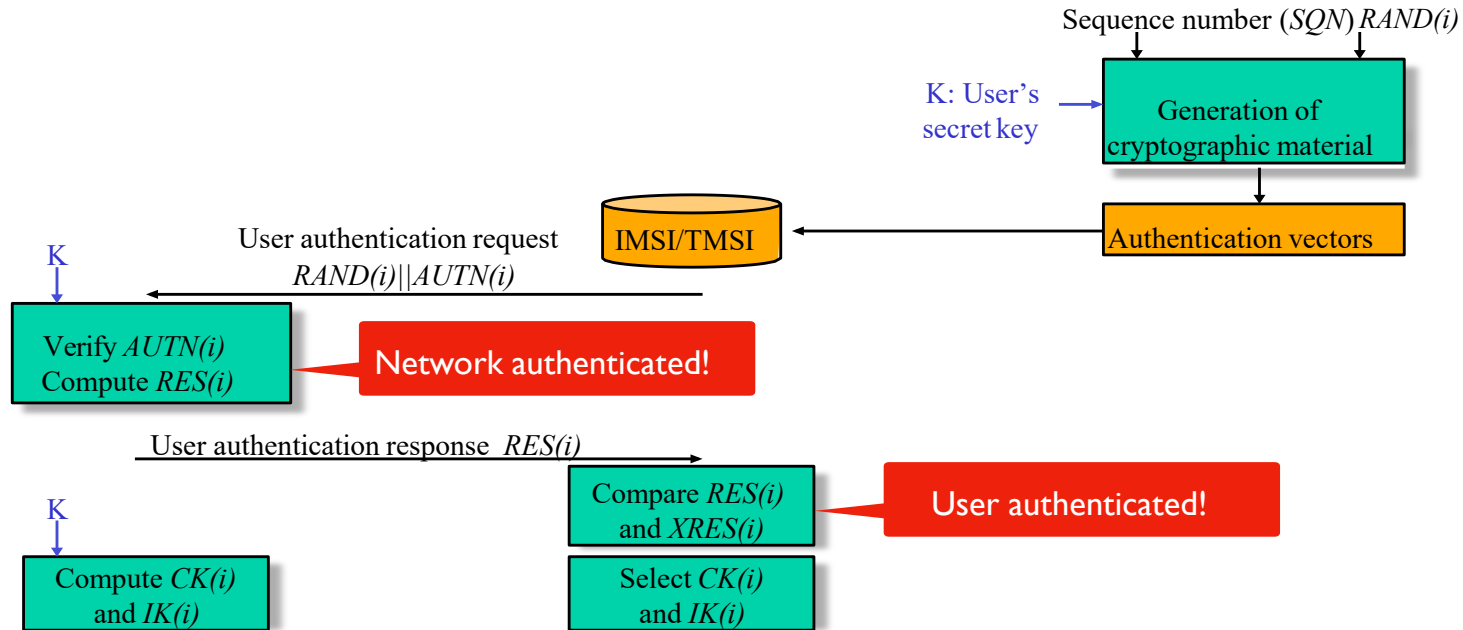
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UMTS Authentication (with a Visited Network)

Mobile Station

Visited Network

Home Environment



Sécurité

1. Le réseau (HLR/AuC) génère les valeurs :

- Il connaît la clé secrète K partagée avec l'UE.
- Il génère :
 - $RAND$ (aléatoire)
 - SQN
 - $AK = f_5(K, RAND)$
 - $MAC = f_1(K, SQN || RAND || AMF)$
 - $AUTN = (SQN \oplus AK) || AMF || MAC$
 - $AUTH = (SQN \oplus AK) || f_5(K, RAND) || f_1(K, SQN || RAND || AMF)$

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Le réseau

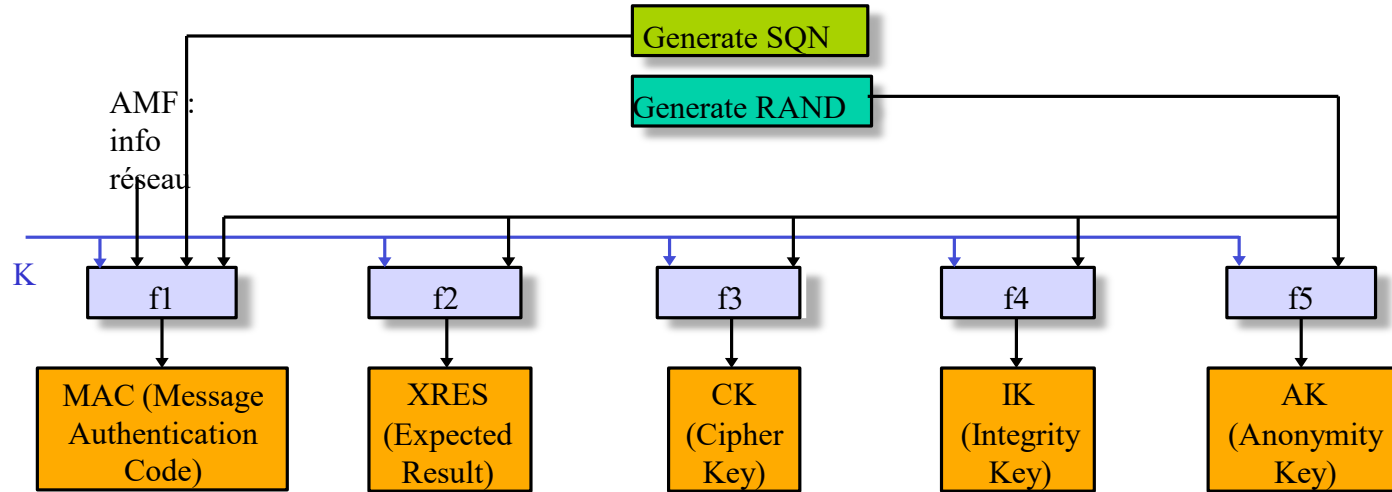
- $AUTN = (SQN \oplus AK) || AMF || MAC$

Définition des composants :

- Élément Signification
- SQN Sequence Number : numéro de séquence utilisé pour éviter les attaques de relecture
- AK Anonymity Key : utilisée pour masquer SQN (protection de la vie privée)
- $\oplus$ OR (ou exclusif) : opération binaire pour rendre SQN illisible
- AMF Authentication Management Field : champ de gestion-2bytes, peut contenir : infos réseau
- MAC Message Authentication Code : tag de sécurité, preuve que le message n'a pas été modifié
- Le réseau envoie : **RAND, AUTN**

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Generation of Authentication Vectors HN



Authentication token: $AuTN = (SQN \oplus AK) || AMF || MAC$

Authentication vector: $AV = RAND || XRES || CK || IK || AuTN$

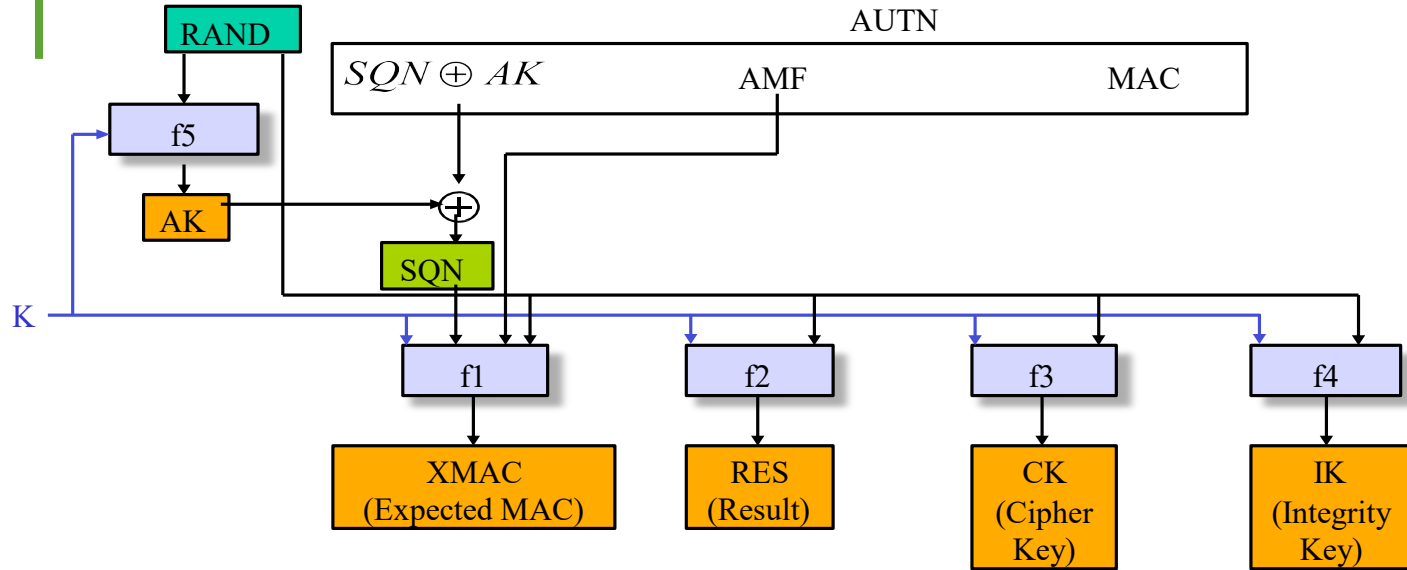
AMF: Authentication and Key Management Field

Le mobile (UE) vérifie le réseau

- Le mobile utilise sa propre clé K pour calculer :
- $AK = f_5(K, RAND)$
- Récupère SQN en démasquant : $(SQN \oplus AK) \oplus AK = SQN$
- Vérifie le MAC avec f_1
- Si MAC correspond, le réseau est authentique.
- Si SQN est valide, le message n'est pas rejoué (protection anti-replay).

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User Authentication in USIM



- Verify $MAC = XMAC$
- Verify that **SQN** is in the correct range

USIM: User Services Identity Module

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Exemple

 $K = 1100\ 1011\ 1010\ 1110\ 1010\ 1110$

- $RAND$ (aléatoire) $0\ 1110$, $SQN\ 8 = 0000\ 0111$, $AMF = 110\ AK$
 $= f_5(K, RAND) = f_5 \oplus (K, RAND) =$
- $MAC = f_1(K, SQN \parallel RAND \parallel AMF) =$
- $AUTN = (SQN \oplus AK) \parallel AMF \parallel MAC$
- $AUTH = (SQN \oplus AK) \parallel f_5(K, RAND) \parallel f_1(K, SQN \parallel RAND \parallel AMF)$

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More About Authentication and Key Generation

In addition to f_1 , f_2 , f_3 , f_4 and f_5 , two more functions are defined: f_1^* and f_5^* , used in case the authentication procedure gets desynchronized (detected by the range of SQN).

f_1 , f_1^* , f_2 , f_3 , f_4 , f_5 and f_5^* are operator-specific

However, 3GPP provides a detailed example of algorithm set, called MILENAGE

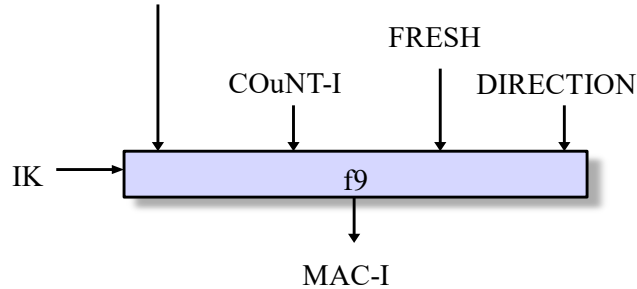
MILENAGE is based on the Rijndael block cipher

In MILENAGE, the generation of all seven functions $f_1 \dots f_5^*$ is based on the Rijndael algorithm

Signaling Integrity Protection

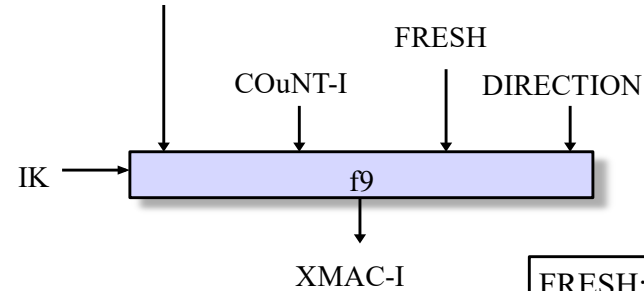
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SIGNALLING MESSAGE



Sender (Mobile Station or Radio Network Controller)

SIGNALLING MESSAGE

Receiver
(Radio Network Controller or Mobile Station)

FRESH: random input

UE → Réseau :

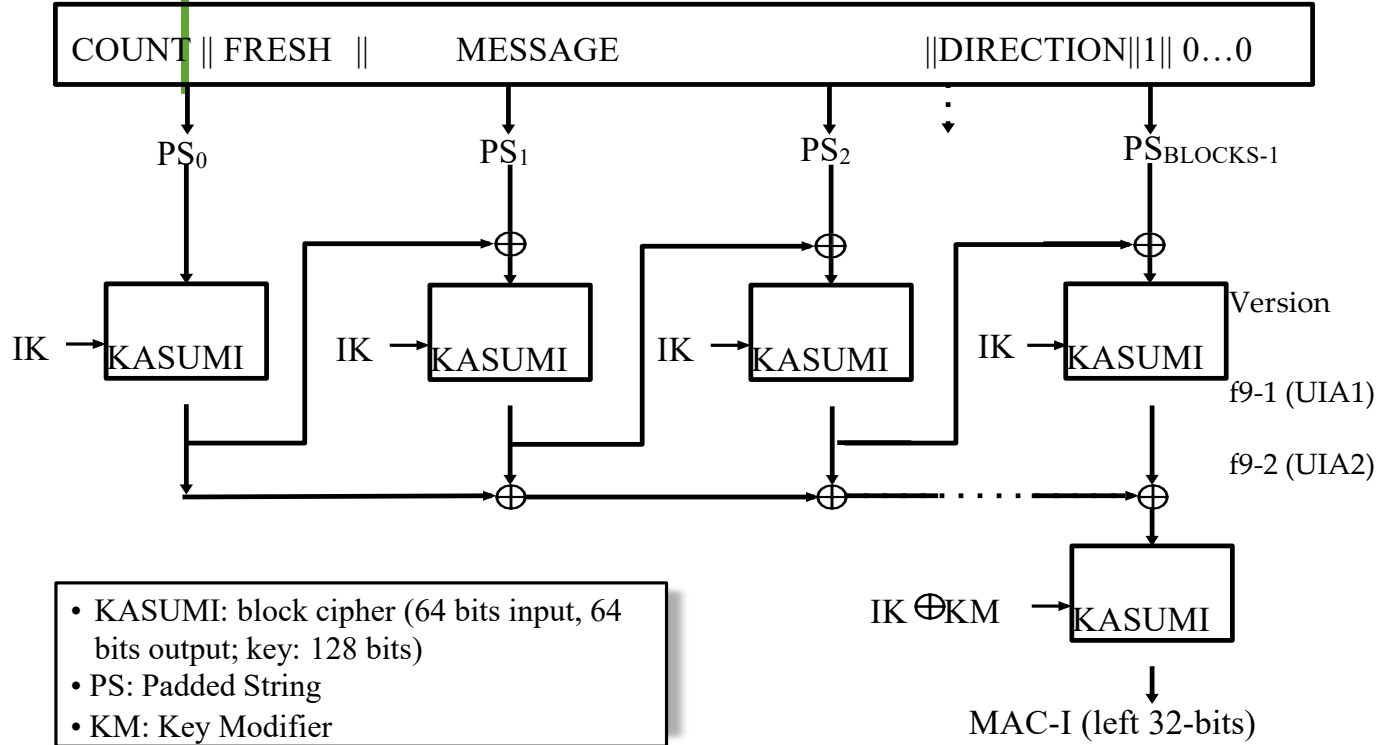
[Message utilisateur] + [MAC-I = $f_9(\text{IK}, \text{message})$]

Réseau :

- Calcule f_9 sur le même message
- Compare les MAC
- Si égaux : le message est accepté

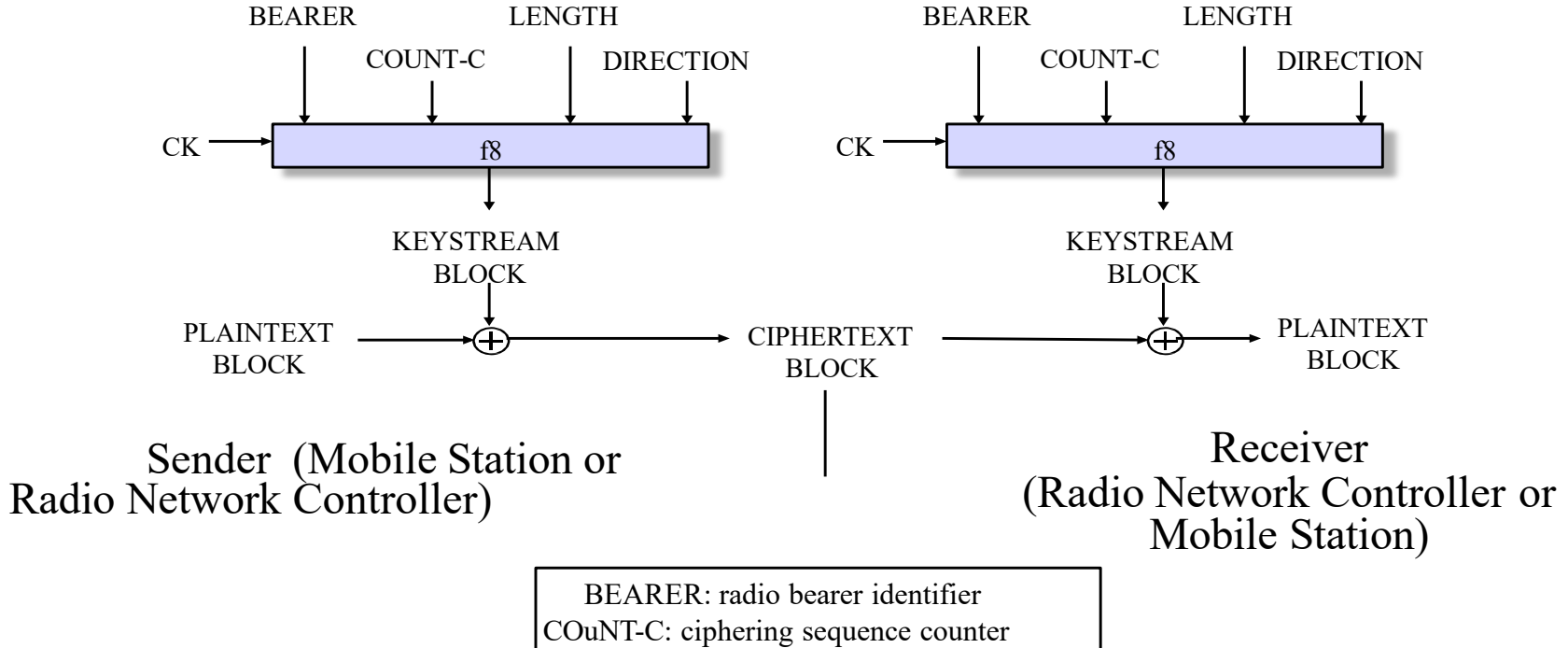
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f9 integrity function



Encryption

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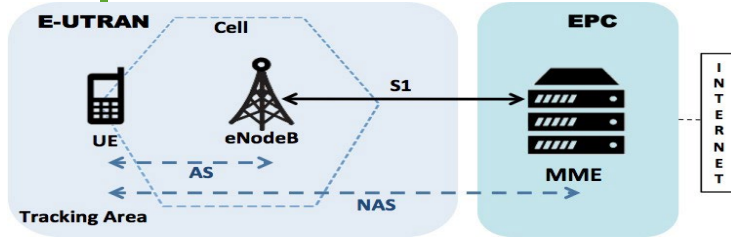
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Long-Term Evolution (**LTE**) **4G**

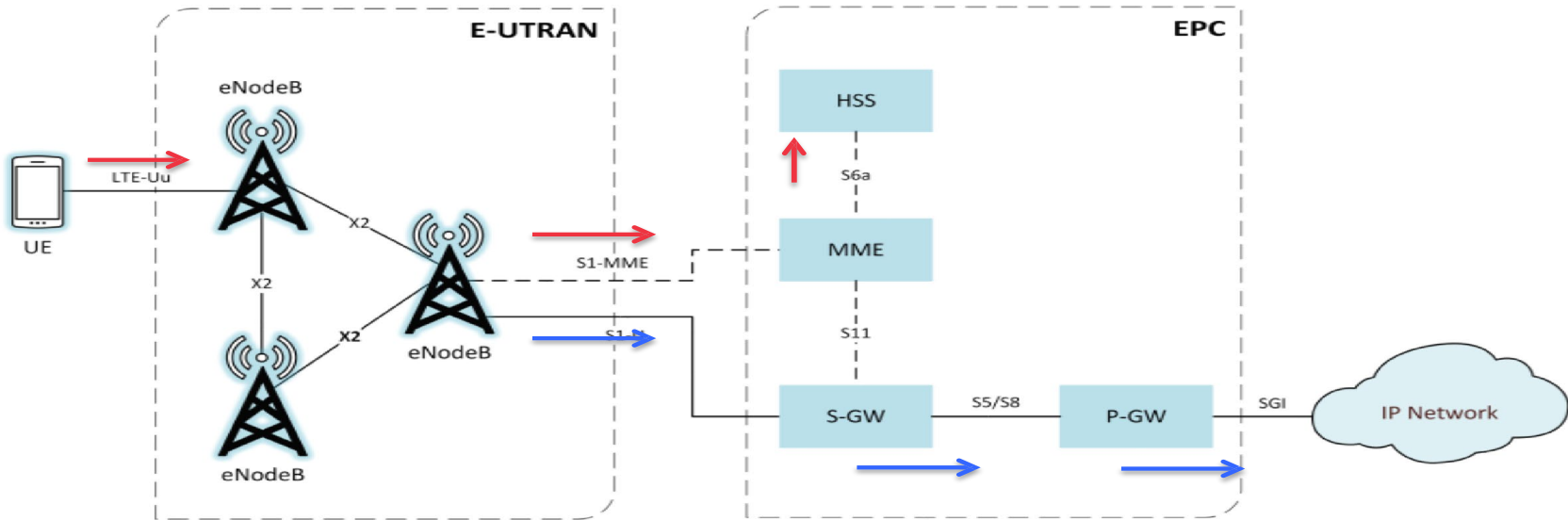
- Re-use of UMTS Authentication and Key Agreement (AKA)
- Extended Key Hierarchy
- Possibility for longer keys
- Greater backhaul protection

LTE System Architecture

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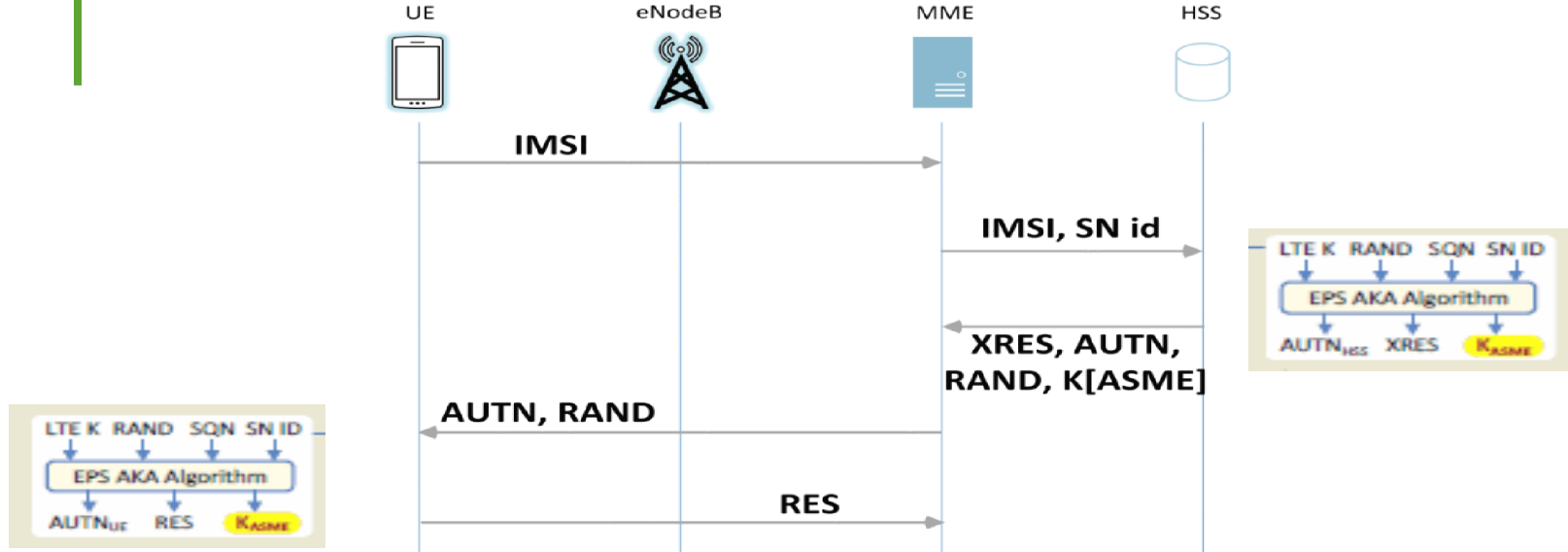


- eNodeB: RF component of LTE
- MME: Mobility Management Entity (Primary Signalling Node)
- HSS: Home subscriber service. Similar to HLR in 2G



LTE (Authentication)

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Network Authentication:
 $\text{AUTN}_{\text{ue}} = \text{AUTN}_{\text{hss}}$

User Authentication:
 $\text{RES} = \text{XRES}$

KASME : Key Access Security Management Entries

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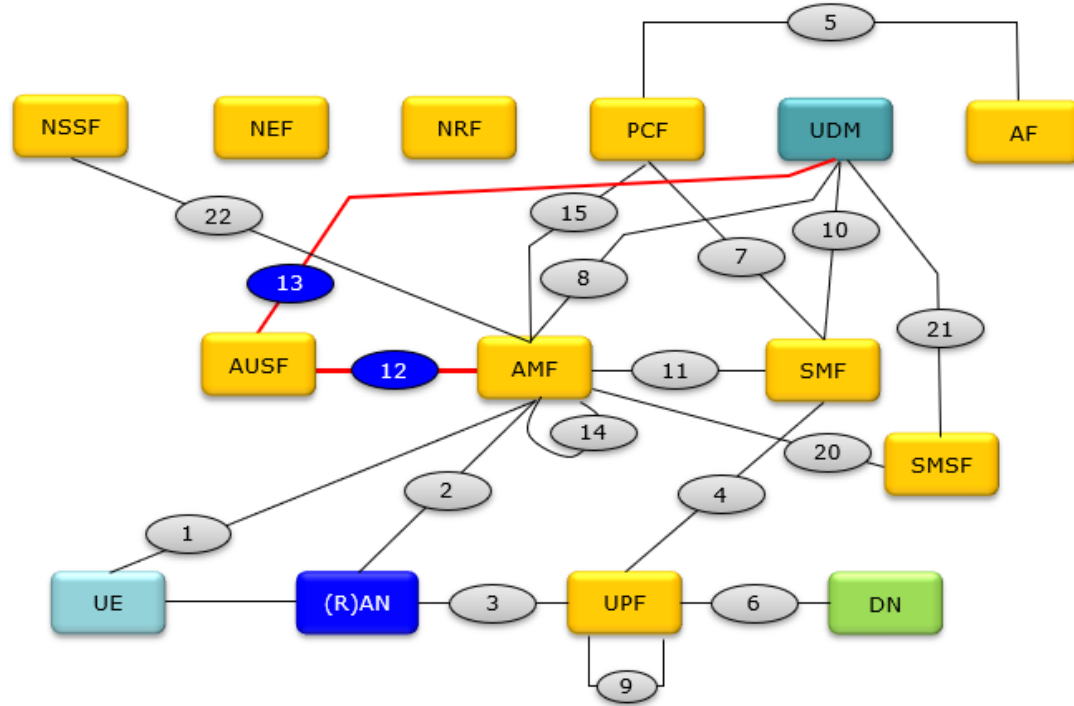
Security Algorithms

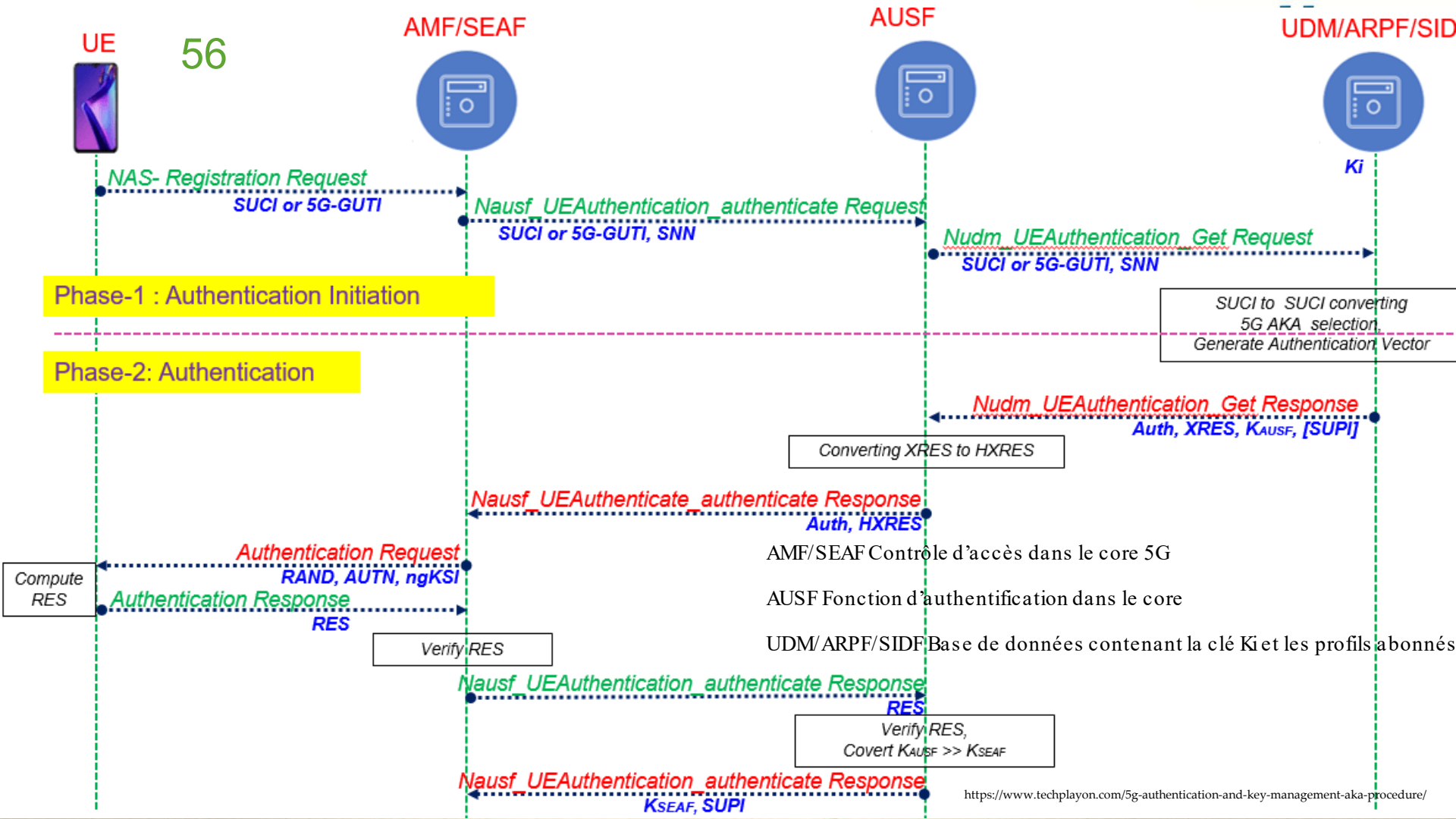
- Two different mandatory 128-bit EPS ciphering and integrity algorithms for CP and UP from day one
 - Snow3G (UMTS based, UIA2 and UEA2) and
 - AES (by US NIST, FIPS standard 197) algorithms
- Algorithm-id:

–	"0000"	128-EEA0	NULL ciphering algorithm
–	"0001"	128-EEA1	SNOW 3G
–	"0010"	128-EEA2	AES
–	"0001"	128-EIA1	SNOW 3G
–	"0010"	128-EIA2	AES

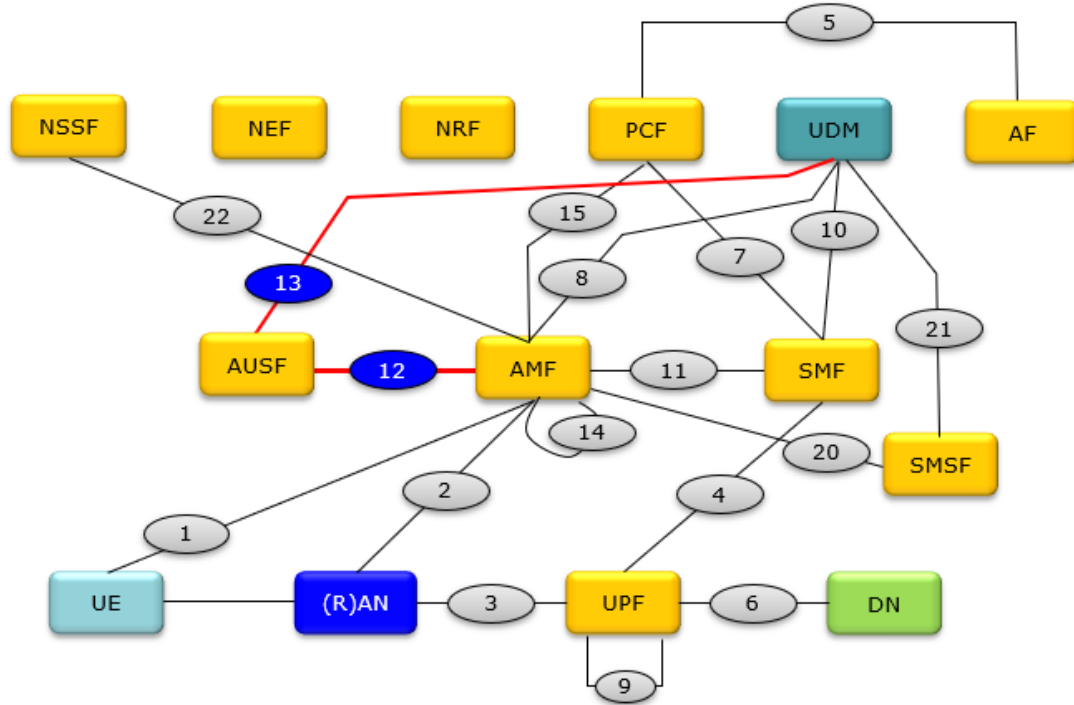
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Sécurité 5G

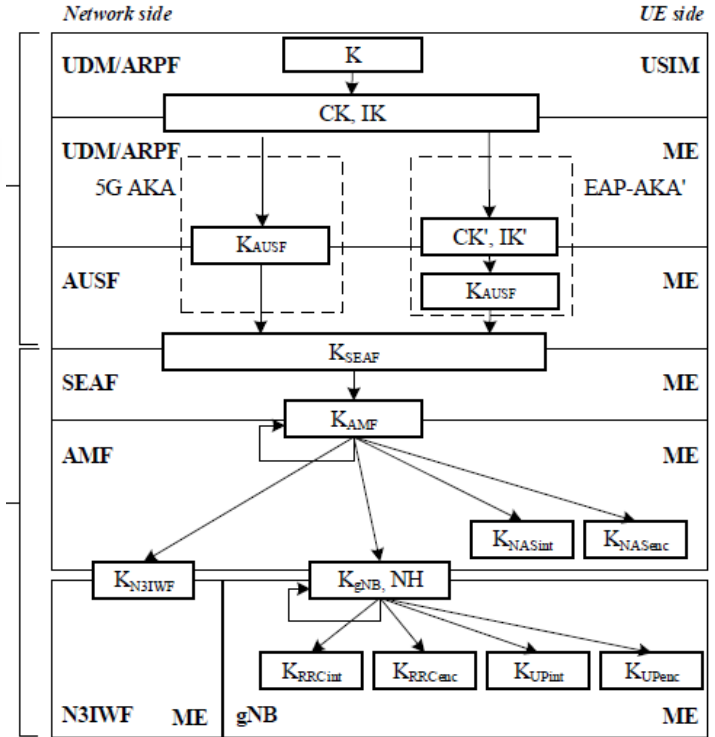




57 | Comparaison



https://www.sharetechnote.com/html/5G/5G_Core_Authentication.html



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Exemple simplifié – Calcul d'un MAC avec AES-CMAC (NIA1)

Objectif : protéger un message avec **intégrité** via AES-CMAC

Données d'entrée :

- Clé d'intégrité : IK = 0x2b7e151628aed2a6abf7158809cf4f3c
- Message (exemple) : "HELLO"
- COUNT = 0x00000001
- BEARER = 0x0A
- DIRECTION = 0 (uplink)

Étapes :

1.Formater les données selon 5G spec (concaténer COUNT, BEARER, DIRECTION, message)

000000010A001001000 01000101 01001100 01001100 01001111

0000000100001010001001000 01000101 01001100 01001100 01001111

2.Passer ces données dans **AES-CMAC** avec la clé IK

3.Résultat : **MAC-I** = 32 bits (ex : 0x6a1f08ed)

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COMPARAISON

Aspect	3G (UMTS)	4G (LTE)	5G (NR)
Fonction d'intégrité	f9 (basé sur KASUMI)	EIAx (UIA1 = SNOW 3G, UIA2 = AES-CMAC)	NIAx (NIA1 = AES-CMAC, NIA2 = ZUC, NIA3 = HMAC-SHA-256)
Fonction de chiffrement	f8 (KASUMI)	EEAx (EEA1 = SNOW 3G, EEA2 = AES)	NEAx (NEA1 = AES, NEA2 = ZUC, NEA3 = SNOW 3G)
Clés utilisées	K (unique)	K + dérivés (eKSI, K_ASME)	K + dérivés (K_AMF, K_gNB, etc.)
Protection signalisation	Oui (f9 sur NAS/RRC)	Oui (EIA sur NAS/RRC)	Oui (NIA sur NAS/RRC/UP)
Protection du plan utilisateur	Non	Oui (EEA sur le plan utilisateur)	Oui (NEA sur le plan utilisateur)
Crypto-agilité / PQ readiness	Non	Moyenne	Forte (préparée au post-quantique)
Algorithmes modulables	Non	Partiellement	Oui (algos négociables dynamiquement)
Anonymat renforcé	Faible (IMSI clair)	Moyen (GUTI)	Fort (SUCI – identifiant chiffré)

References

- **AuthLoop: End-to-End Cryptographic Authentication for Telephony over Voice Channels**, B. Reaves, L. Blue and P. Traynor, In Proceedings of the USENIX Security Symposium (SECURITY), 2016.
- **SS7: Locate. track. manipulate.** Engel, Tobias, 31st Chaos Computer Congress.

Recommended Readings

- **On Security Research towards Future Mobile Network Generations**, David Rupprecht, Adrian Dabrowski, Thorsten Holz, Edgar Weippl, and ChrisCna Poepper, <https://arxiv.org/pdf/1710.08932.pdf>
- **Security for Telecommunications Networks. Springer, Series: Advances in Information Security**, P. Traynor, P. McDaniel and T. La Porta, August, 2008. (available ETH online library)
- NIST SP 800-187 GUIDE TO LTE SECURITY